

SECTION 1 PREVENTIVE MAINTENANCE SCHEDULE

IR-7000 Preventive Maintenance Manual

Educational Sample Document.

Daily checks:

- Inspect cable dress pack for abrasion.
- Verify teach pendant cable strain relief.
- Confirm cooling fan intake is free of debris.
- Review alarm history for recurring faults.

Monthly checks:

- Check grease purge ports for fresh grease indication on J2 and J3.
- Inspect brake operation during controlled motor-off drop test in Teach.
- Tighten base mounting bolts to 85 N·m.
- Clean controller cabinet filters.

Annual checks:

- Replace controller cabinet filter elements.
- Inspect harmonic drive seals on wrist axes.
- Measure motor insulation resistance (minimum 10 M $\cdot$ at 500 VDC).
- Backup controller parameters and programs to external media.

SECTION 2 COOLING SYSTEM

The RC-7000 controller uses forced-air cooling. Two axial fans draw air through the front filter and exhaust at the rear.

Symptoms of cooling problems:

- Cabinet overtemperature warning A-305
- Unexpected thermal derating of servo amplifiers
- Elevated measured cabinet temperature above 45°C ambient inside enclosure

Recommended checks:

1. Clean or replace the intake filter.
2. Verify both fans rotate when control power is on.
3. Ensure 100 mm clearance behind the cabinet for exhaust.
4. Confirm ambient room temperature remains within 5-40°C.

If A-305 persists after filter and fan checks, inspect temperature sensor TS1 on the amplifier backplane. Do not operate with blocked ventilation.

SECTION 3 LUBRICATION

Grease specification: Lithium complex EP grease, NLGI grade 2, robot grade.

J2 and J3 reducers: purge 8-10 grams every 6 months under rated duty.

Wrist axes J4-J6: inspect seals; grease only if manufacturer purge ports are present.

Over-greasing can raise seal pressure and cause leaks. Wipe excess grease.

Dispose of waste grease according to local regulations.

After lubrication, jog each axis through full travel in Teach at reduced speed.

SECTION 4 BATTERY AND MEMORY

The controller encoder backup battery maintains absolute encoder data when main power is removed.

Battery replacement interval: every 24 months, or immediately if alarm A-120 (encoder battery low) appears.

Procedure:

1. Place the robot in a stable posture and note the position.
2. Turn main power OFF and apply lockout/tagout.
3. Replace battery BATT1 on the controller main board within 5 minutes to reduce risk of encoder data loss.
4. Restore power and verify no A-121 (encoder data lost) alarm.
5. If A-121 is present, remaster axes per the Calibration section.

Always follow site lockout/tagout procedures before opening the cabinet.